TOUCH GRASS

AI-Powered Real-World Challenge Game

Tagline:

The phone is the scoreboard. The game happens outside.

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1. Product Overview

Touch Grass is a real-world competitive game where players receive personalized outdoor challenges generated by AI.

The player chooses the kind of experience they want — such as **Calm, Explore, Social, Active, Observe, or Surprise** — and the AI Game Master generates a challenge based on their choice, available time, context, and previous activity.

The player then goes outside and completes the challenge.

Depending on the challenge, they may need to:

- walk or move for a certain amount of time
- explore somewhere unfamiliar
- find something unusual
- take a photograph
- observe something
- remember something and answer questions afterward
- complete an activity with friends
- reach a particular type of location

After completing the challenge, the player returns to the app and submits whatever evidence the challenge requires.

The game awards a **baseline XP reward**, and AI can award additional bounded bonuses based on the quality, relevance, novelty, or creativity of the player's submission.

Players can also create **Spaces**, invite friends using their Touch Grass IDs, and complete challenges together.

The central loop is:

Choose → Generate → Go Outside → Complete → Prove → Earn XP → Compete

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2. The Core Product Idea

Touch Grass should NOT feel like:

- a fitness tracker
- a walking app
- a social media platform
- an AI chatbot
- a navigation app
- a photo-sharing application

Those are supporting mechanisms.

The product is:

A game that generates things for you to actually do in the real world.

The AI is the **Game Master**.

The phone is primarily used to:

- choose a challenge
- start the challenge
- provide occasional evidence
- submit evidence afterward
- track progress
- see scores and rankings

The user should spend significantly more time **doing the challenge outside** than interacting with the application.

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3. The Problem

Modern apps are designed to keep people looking at screens.

Even apps intended to encourage outdoor activity often reduce the experience to:

- steps
- distance
- calories
- maps
- streaks
- statistics

Touch Grass treats going outside as the **game itself**.

Instead of telling someone:

"Walk for 5,000 steps."

it can say:

"Take a route you've never taken. Find something you've probably walked past before but never noticed. Photograph it."

The objective is not simply physical activity.

It is **real-world engagement, exploration, novelty, and social interaction.**

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4. Target Users

Primary audience:

- teenagers
- university students
- young adults
- people who spend a lot of time online
- people who want something fun to do outside
- friend groups looking for activities
- people who enjoy games, challenges, XP and competition

The game should work for someone who is:

- completely alone
- with one friend
- with several friends
- in a familiar neighborhood
- in a new environment

No organizer, physical clue placement, QR codes, stickers, or special infrastructure should be required.

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5. The Two Ways to Play

A. Solo

The player generates a challenge for themselves.

Example:

HIDDEN IN PLAIN SIGHT

Take a route you don't normally take.

Find something you've probably walked past before but never actually noticed.

Take one photograph.

Base reward: 500 XP

Discovery bonus: up to +300 XP

The player completes it independently.

B. With Friends

Players can create a **Space**.

A Space is a temporary or persistent shared game room for a group.

Example:

Saturday Side Quests

Rebecca
Daniel
Maya
James

The creator can invite people using their **Touch Grass ID**.

The Space can generate challenges that everyone in the group can complete.

Example:

WHO CAN FIND THE STRANGEST THING?

Everyone has 30 minutes.

Go outside and find something genuinely unusual.

Take one photo.

Return and submit your discovery.

Base: 500 XP

Discovery bonus: up to +300 XP

Everyone goes outside.

They don't need to stay inside the app.

When everyone returns, the submissions are evaluated and scores are shown.

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6. The Experience Selection

When generating a challenge, the player sees:

What do you feel like doing?

🌿 Calm

Something slower and more peaceful.

□ Explore

Discover somewhere or something unfamiliar.

□🤝□ Social

An activity designed around friends or interaction.

🏃 Active

More movement-focused.

👀 Observe

Pay attention to things people normally overlook.

📸 Discover

Find something interesting and document it.

🎲 Surprise Me

Let the AI decide.

The player can also specify:

How much time do you have?

- 10 minutes
- 20 minutes
- 30 minutes
- 1 hour

Who are you with?

- Alone
- One friend
- Several friends

The system can also consider:

- previous challenges
- previous challenge types
- difficulty
- recent activity
- player preferences
- repetition
- current Space context

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7. Example Generated Challenges

Calm

THE QUIET DETOUR

You have 20 minutes.

Take a route you've never taken before.

Find somewhere you can sit comfortably for five minutes.

Put your phone away.

Pay attention to three sounds you normally ignore.

When you return, tell Touch Grass what you noticed.

Base: 450 XP

Explore

UNKNOWN TERRITORY

Leave your usual route.

Find somewhere you've never been before.

It could be a street, shop, building, park, or other interesting place.

Take one photo of something that caught your attention.

When you return, explain why you chose it.

Base: 600 XP

Observe

THE THING YOU NEVER NOTICE

Walk somewhere familiar.

Find one thing you've probably passed many times but never actually noticed.

Take a photo.

Explain what made you notice it this time.

Base: 500 XP

Active

THE DETOUR

Walk continuously for 20 minutes.

You cannot take your usual route.

Whenever you reach a familiar decision point, choose the option you normally ignore.

Base: 650 XP

Social

THE SIDE QUEST

You and your friend have 30 minutes.

Take turns choosing which direction to walk whenever you reach an intersection.

You cannot use navigation.

Your goal is to find somewhere neither of you has visited before.

Take one photo together.

Base: 750 XP

Surprise

THE WRONG TURN

For the next 15 minutes, you are not allowed to follow your normal route.

At every decision point, choose the option you would normally ignore.

Find one thing you have never noticed before.

Base: 600 XP

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8. The Evidence System

Not every challenge should require the same kind of proof.

Each generated challenge has a **verification method**.

Possible methods include:

📸 Photo

The player must take a photograph during the activity.

🚶 Activity

The system checks movement, duration, distance, or steps where available.

📍 Location

The system checks whether the player reached an appropriate area.

🧠 Memory

The player must remember something encountered outside and answer questions when they return.

👀 Observation

The player submits a description or photograph of what they discovered.

🧑‍🤝‍🧑 Social

The challenge can require multiple members of a Space to participate.

🍃 Honor System

Some experiences, particularly calm or reflective activities, cannot meaningfully be objectively verified.

This is acceptable.

The goal is not to create a surveillance system.

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9. Photography

Photography is an important mechanic but **not the entire product**.

The photo must be taken **during the real-world activity** when the challenge requires one.

For example:

Find something unexpectedly red during your walk.

Take one photograph.

The user goes outside, finds something, and takes the photo.

The photograph becomes evidence.

The app can then use AI to determine whether the submitted image is relevant to the generated challenge.

Photography can also create a personal collection:

My Discoveries

Examples:

- weird architecture
- unusual signs
- hidden artwork
- strange objects
- interesting plants
- unexpected places

This can become a secondary reason to return to the app.

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10. Scoring System

This is a key part of Touch Grass.

Every challenge has a **deterministic baseline XP reward**.

AI does NOT decide the entire reward.

For example:

Challenge baseline: **500 XP**

Gemma evaluates the evidence and can award a bounded bonus.

Possible bonus categories:

- Relevance
- Novelty
- Interestingness
- Creativity
- Quality of explanation
- Challenge difficulty
- Evidence quality

Example:

Base

500 XP

Gemma evaluation

Relevance: +80
Novelty: +120
Interestingness: +60

Final

760 XP

The AI bonus must have a hard maximum.

For example:

Base reward = 500 XP
Maximum AI bonus = +300 XP
Maximum total = 800 XP

The AI cannot arbitrarily give someone 2,000 XP.

The deterministic game engine calculates the final score.

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11. Example: Rare Discovery

Challenge:

FIND SOMETHING RARE

During your walk, find something you don't normally encounter.

Take one photograph.

Base: 500 XP

Potential discovery bonus: **+300 XP**

Player A finds a normal parked car.

Gemma:

- relevance: low
- novelty: very low
- interestingness: low

Bonus:

+30 XP

Total:

530 XP

Player B finds an extremely unusual old sign.

Gemma:

- relevance: high
- novelty: high
- interestingness: high

Bonus:

+270 XP

Total:

770 XP

The important distinction:

Gemma should not claim:

"This is objectively the rarest thing in your city."

Instead, it should evaluate:

"How well does this submission satisfy the challenge?"

This keeps the scoring defensible.

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12. Friend Competition

Friend competition should be lightweight.

Players can see:

- current XP
- level
- completed challenges
- streak
- ranking

Example:

FRIENDS

#1 Maya — 8,420 XP
#2 Rebecca — 7,980 XP
#3 Daniel — 7,640 XP

Maya is only 180 XP ahead.

The goal is to create a reason to come back without turning Touch Grass into another social media feed.

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13. Spaces

Spaces are shared challenge environments.

A user can:

1. Create a Space.
2. Give it a name.
3. Invite friends using their Touch Grass IDs.
4. Generate a group challenge.
5. Everyone completes it outside.
6. Everyone submits evidence.
7. Results are calculated.
8. XP is awarded.

Example:

Saturday Side Quests

4 players

ACTIVE CHALLENGE

Find the most unexpected thing you encounter during a 30-minute walk.

Submit one photo.

Base: 500 XP

Bonus: up to +300 XP

After completion:

RESULTS

🥇 Maya — 820 XP
🥈 Rebecca — 760 XP

🥉 Daniel — 690 XP
James — 540 XP

This creates an actual game rather than merely a friend leaderboard.

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14. The Player Scenario

This is the intended experience.

Scenario: Rebecca Plays Solo

Rebecca opens Touch Grass.

Her dashboard says:

LEVEL 8

3,420 XP
🔥 5-day streak

What do you feel like doing?

She chooses:

EXPLORE

Then:

How much time do you have?

20 minutes

She chooses:

Alone

Touch Grass sends this request through its AI Game Master workflow.

Gemma generates:

HIDDEN IN PLAIN SIGHT

Take a route you don't normally take.

Find something you've probably walked past before but never actually noticed.

Take one photograph of it.

When you return, explain why it stood out to you.

Base reward: 500 XP

Discovery bonus: up to +300 XP

Rebecca taps:

START CHALLENGE

The app tells her:

You've got everything you need.

Go.

She leaves her phone in her pocket.

She walks.

After about ten minutes, she notices an old hand-painted sign on a building that she's walked past many times.

She takes one photograph.

Then she continues her walk.

She returns home and opens Touch Grass.

The app asks:

What made you notice this?

She writes:

"I've passed this building many times but never noticed the old lettering."

She submits.

The system checks:

- challenge requirements
- photo
- explanation
- available activity/location evidence

Gemma evaluates the discovery:

Relevance: high
Novelty: high
Explanation: strong

Discovery bonus: +240 XP

The game engine calculates:

Base: 500 XP
AI bonus: +240 XP

TOTAL: 740 XP

Then:

NEW RANK

#187 Global

You passed Daniel.

Rebecca can now:

Play Again

or:

Create a Space

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15. Scenario: Playing With Friends

Rebecca creates:

SATURDAY SIDE QUESTS

She receives a Space ID.

She adds:

- Maya
- Daniel
- James

using their Touch Grass IDs.

She selects:

Social

and:

30 minutes

The Game Master creates:

THE STRANGEST FIND

Everyone has 30 minutes.

Go somewhere outside your normal route.

Find the most unusual thing you encounter.

Take exactly one photograph.

Return and submit your discovery.

Base: 500 XP

Discovery bonus: up to +300 XP

Everyone leaves the app.

They go outside.

Maya finds an unusual piece of architecture.

Rebecca finds an old sign.

Daniel finds an unusual object.

Everyone submits.

Gemma evaluates each submission using the **same scoring rules**.

The Space displays:

RESULTS

🥇 Maya — 820 XP
🥈 Rebecca — 760 XP
🥉 Daniel — 690 XP
James — 580 XP

Maya found today's most interesting discovery.

The group can generate another challenge.

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16. The AI Game Master

The AI should have two major responsibilities.

Responsibility 1: Generate Challenges

Gemma receives:

- selected activity type
- available time
- solo/group context
- player preferences
- recent challenges
- previous challenge types
- difficulty
- Space information
- optional environment information

It returns a structured challenge.

Example:

```
{
  "type": "exploration",
  "title": "Hidden in Plain Sight",
  "description": "Find something you have probably walked past before but
never noticed.",
  "duration_minutes": 20,
  "verification": ["photo", "text_response"],
  "base_xp": 500,
  "max_bonus_xp": 300,
  "difficulty": "medium"
}
```

The backend validates this response before the challenge becomes playable.

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17. AI Responsibility 2: Evaluate Evidence

After a challenge is completed, Gemma can evaluate submitted evidence.

For a photo challenge, it can evaluate:

- Does the photo appear relevant?

- Does it satisfy the challenge?
- Is the discovery genuinely interesting relative to the challenge?
- Is the explanation consistent with the image?
- How strong is the submission?

It returns a structured evaluation.

Example:

```
{
  "valid": true,
  "relevance_score": 92,
  "novelty_score": 81,
  "interestingness_score": 88,
  "bonus_xp": 240
}
```

The game engine then calculates the final score.

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18. Important AI Boundary

The AI should **never directly control the game's authoritative state.**

Gemma can:

- propose a challenge
- evaluate evidence
- recommend a bonus

The backend/game engine decides:

- whether the challenge is valid
- whether requirements were satisfied
- the final XP
- leaderboard position
- level
- achievements
- whether a player can claim a reward

This prevents an LLM from accidentally breaking the game's economy.

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19. Gemma

Role: AI Game Designer + Evidence Judge

Gemma is the primary intelligence behind the experience.

It is used for:

Challenge generation

Personalized outdoor activities.

Challenge variation

Avoiding repetitive missions.

Evidence evaluation

Understanding submitted photos/text.

Discovery scoring

Giving bounded bonuses based on the player's submission.

Personalization

Learning what types of challenges a player enjoys.

Without Gemma, Touch Grass would mostly be a fixed challenge database.

With Gemma, the game can continuously create new experiences.

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20. Mastra

Role: AI/Game orchestration layer

Mastra manages the workflow around Gemma.

Conceptually:

```
Player Request
      ↓
Retrieve Player History
      ↓
Determine Constraints
      ↓
Mastra Workflow
      ↓
Gemma
      ↓
Structured Challenge
      ↓
```

```
Validation
      ↓
Store Challenge
      ↓
Player Completes It
      ↓
Evidence Submitted
      ↓
Mastra/Gemma Evaluation
      ↓
Game Engine
      ↓
XP + Leaderboard
```

Mastra should not exist merely to say the project uses Mastra.

It should actually orchestrate the Game Master workflow.

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21. MongoDB Atlas

Role: Persistent game state and player memory

MongoDB Atlas stores:

Users

- Touch Grass ID
- username
- level
- XP
- streak
- preferences

Challenges

- generated challenge
- type
- difficulty
- requirements
- base XP
- bonus limit
- verification method

Completions

- player
- challenge
- evidence
- completion status

- AI evaluation
- XP awarded

Spaces

- Space ID
- members
- challenges
- results

Player history

- completed challenge types
- favorite experiences
- previous discoveries
- recent challenges

This persistent history allows the AI to avoid generating the same experience repeatedly.

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22. ElevenLabs

Role: Optional Game Master voice

ElevenLabs should be used selectively.

It should NOT constantly talk to the player.

Instead, voice can be used for special experiences.

Example:

"For the next twenty minutes, leave your normal route behind. When you find somewhere you've never sat before, put your phone away..."

The player can start the audio and lock the phone.

This makes ElevenLabs directly support the central idea:

The phone should facilitate the experience rather than become the experience.

Potential uses:

- special challenges
- calm challenges
- event introductions
- level-up moments

- rare challenges

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23. Temporal

Role: Durable scheduled game workflows

Temporal can be used for longer-running game processes such as:

- daily challenge generation
- weekly challenges
- Space events
- streak management
- weekly competitions
- challenge deadlines
- event closing
- reward calculation

Example:

```
Start Weekly Event
        ↓
Generate Challenge
        ↓
Event Active
        ↓
Players Complete Challenges
        ↓
Deadline
        ↓
Calculate Results
        ↓
Award Rewards
        ↓
Close Event
```

Temporal is optional if time becomes a constraint.

It is better to have a smaller number of genuinely useful integrations than to add technology purely for the sake of mentioning it.

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24. Partner Technology Summary

Technology	What it does
Gemma	Generates personalized challenges and evaluates evidence
Mastra	Orchestrates the AI Game Master workflow
MongoDB Atlas	Stores users, challenges, history, Spaces and game state

Technology	What it does
ElevenLabs	Provides optional voice-based challenge experiences
Temporal	Handles durable scheduled events/workflows
Next.js + TypeScript	Frontend/application
Tailwind CSS	UI styling
Backend/Game Engine	Validation, XP, scoring, leaderboards
Activity/GPS APIs	Verification where appropriate

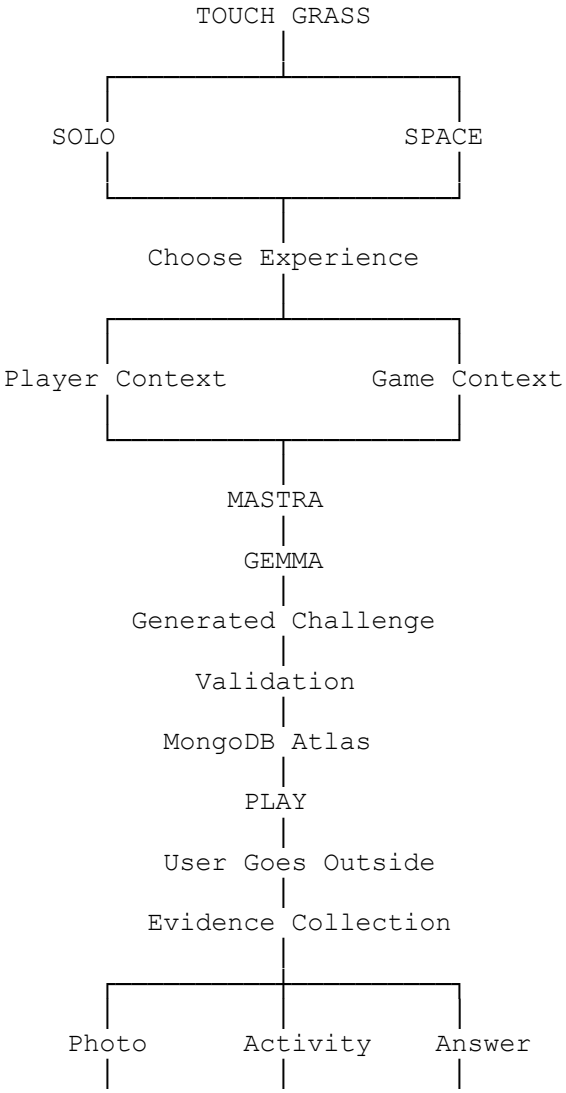
The strongest integrations are:

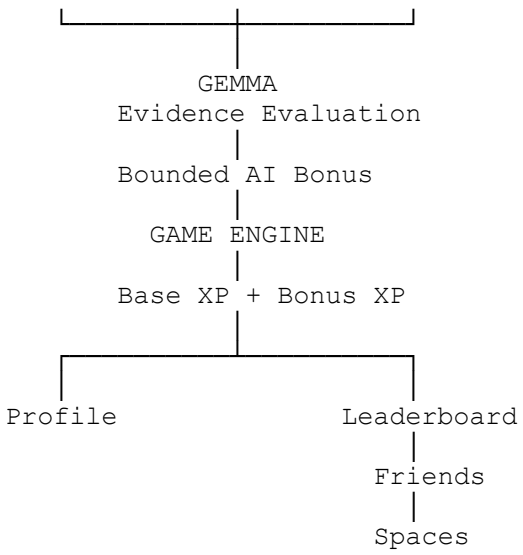
Gemma + Mastra + MongoDB Atlas

ElevenLabs and Temporal should be added where they genuinely improve the experience.

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25. Game Architecture





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26. XP Philosophy

XP should **not** simply represent steps.

A player should be rewarded for engaging with the game.

XP can come from:

- completing challenges
- exploration
- social challenges
- movement challenges
- successful evidence
- discovery quality
- streaks
- special events
- challenge difficulty

Steps are only one possible input.

This prevents Touch Grass from becoming another step counter.

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27. Verification Philosophy

The system should not try to prove everything with invasive tracking.

Instead:

Use the lightest verification method that makes sense for the challenge.

For example:

Challenge	Verification
Walk 20 minutes	Activity
Reach unfamiliar area	Location
Find unusual object	Photo + AI
Remember something	Post-walk answer
Social challenge	Multiple participants
Calm experience	Mostly honor system
Explore somewhere new	Location/photo
Find something interesting	Photo + AI

This keeps the experience enjoyable.

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28. Anti-Cheat Philosophy

Touch Grass is a game, but it does not need military-grade anti-cheat.

The backend should prevent obvious manipulation of:

- XP
- challenge completion
- leaderboard scores
- duplicate submissions

The AI should not be trusted with authoritative game state.

For important challenges, the app can combine:

activity + timestamp + location + evidence

where appropriate.

But the game should accept that some experiences are inherently subjective.

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29. Main Screens

Home

Show:

- level
- XP
- streak
- ranking
- current Space activity
- start challenge button

Example:

LEVEL 8

3,420 XP
🔥 5-day streak
#187 Global

What do you feel like doing?

Explore | Calm | Social | Active | Observe | Surprise

Challenge Screen

Show:

- title
- instructions
- duration
- verification method
- base XP
- maximum bonus
- Start button

Example:

HIDDEN IN PLAIN SIGHT

Find something you've probably walked past before but never noticed.

📸 Photo required

20 minutes

500 XP + up to 300 bonus

START

Active Challenge

Keep this screen extremely minimal.

CHALLENGE ACTIVE

14:32 remaining

Your mission:

Find something you've never noticed before.

Put your phone away.

[End Challenge]

The player should not need to stare at the screen.

Completion

WELCOME BACK

Submit your discovery.

[Upload Photo]

What made you choose this?

[Text box]

SUBMIT

Results

CHALLENGE COMPLETE

Base XP: +500
Discovery bonus: +240

TOTAL: +740 XP

🏆 New rank: #187

You passed Daniel.

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30. Profile

Show:

- username
- Touch Grass ID
- level
- total XP
- streak
- completed challenges
- favorite challenge types
- discoveries
- achievements

Potential section:

MY DISCOVERIES

A visual collection of interesting things the player has found.

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31. MVP

The MVP should focus on the core experience.

Required

- Account/profile
- Touch Grass ID
- Experience selection
- Time selection
- AI challenge generation
- Structured challenges
- Base XP
- Photo submission
- AI evidence evaluation
- bounded AI bonus
- XP calculation
- basic leaderboard
- Spaces
- invite friends by ID
- group challenges
- MongoDB persistence

Nice-to-have

- activity tracking
- location verification
- streaks
- achievements
- discovery collection
- ElevenLabs
- Temporal
- advanced friend rankings

Do not prioritize

- social feed
- chat
- complicated guild system
- extensive maps
- navigation
- dozens of achievement types
- elaborate messaging
- complex fitness analytics

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32. The Product's Competitive Advantage

The outdoor challenge space already contains many projects where:

AI generates something interesting for you to notice outside.

Touch Grass should therefore not be positioned as simply:

"AI generates outdoor quests."

Its stronger positioning is:

Touch Grass is a persistent real-world game where AI designs personalized challenges, players complete them offline, evidence can be verified, and the quality of what they discover can affect their score.

The combination is important:

1. Personalized challenge generation

The user chooses what kind of experience they want.

2. Multiple types of real-world interaction

The game isn't just walking.

3. Evidence

The player can actually demonstrate what they did.

4. AI-evaluated discovery

Gemma can award bounded bonuses based on the quality of the player's submission.

5. Persistent progression

XP, levels, streaks and rankings give the experience continuity.

6. Social competition

Friends can create Spaces and play together.

7. Phone-away philosophy

The application deliberately pushes the user away from the screen.

This combination makes the product feel like a **game**, rather than an AI outdoor assistant.

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33. The One-Sentence Explanation

If a judge asks:

"What does Touch Grass do?"

Answer:

Touch Grass is a real-world challenge game where you choose the kind of experience you want, AI creates a challenge for you, you go outside and complete it, then return to prove it and earn XP. You can play alone or create Spaces to compete with friends.

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34. The Short Pitch

Touch Grass turns going outside into a game.

Choose how you want to spend your time outside, and an AI Game Master creates a personalized challenge for you.

Then put your phone away and actually do it.

When you return, submit the evidence the challenge requires. You earn a guaranteed baseline XP reward, while AI can award additional bonuses for interesting or impressive discoveries.

Play solo or create a Space, invite friends by ID, and compete to see who can actually touch the most grass.

The phone is the scoreboard. The game happens outside.

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35. Core Product Principle

Every feature should answer one question:

Does this make the real-world experience more fun, more meaningful, or more competitive?

If it doesn't, don't build it.

The product should always return to:

Choose → Generate → Go Outside → Complete → Prove → Earn → Compete.

That is Touch Grass.